Assignment
Chapter 1

1. Define an algorithm. List and explain its key properties.
2. What is the RAM model? Define Time and Space Complexity.
3. Explain Asymptotic Notations with it types with its mathematical definition & graph.
4. Solve the following recurrence relation using Recursion Tree Method:
   $T(1)=1$, for $n=1$
   $T(n) = T(n/2) + 1$, for $n>1$
5. Solve the following recurrence relation using Recursion Tree Method:
   $T(1)=1$, for $n=1$
   $T(n) = T(n-1) + 1$, for $n>1$
6. Solve the following recurrence relation using Recursion Tree Method:
   $T(1)=1$, for $n=1$
   $T(n) = 2\,T(n/2) + 1$, for $n>1$
7. Solve the following recurrence relation using Recursion Tree Method:
   $T(1)=1$, for $n=1$
   $T(n) = 2\,T(n/2) + n$, for $n>1$
8. Solve the following recurrence relation using Recursion Tree Method:
   $T(1)=1$, for $n=1$
   $T(n) = T(n/2) + T(n/3) + 1$, for $n>1$
9. Solve the following recurrence relation using Recursion Tree Method:
   $T(1)=1$, for $n=1$
   $T(n) = T(n/4) + T(n/2) + n^2$, for $n>1$
10. Solve the following recurrence relation using Recursion Tree Method:
    $T(1)=1$, for $n=1$
    $T(n) = 2\,T(n/2) + n^2$, for $n>1$
11. Show the $O(n^3)$ is the complexity of the following recurrence relation by using Substitution method.
    $T(1)=1$, for $n=1$
    $T(n) = 4\,T(n/2) + n$, for $n>1$
12. Show the $O(n^2)$ is the complexity of the following recurrence relation by using Substitution method.
    $T(1)=1$, for $n=1$
    $T(n) = 4\,T(n/2) + n$, for $n>1$
13. Show the $O(n^3)$ is the complexity of the following recurrence relation by using Substitution method.

$T(1)=1$, for $n=1$

$T(n) = 8\,T(n/2) + n^2$, for $n>1$

14. Solve the following recurrence relation using Masters Method:
    a) $T(n) = 3\,T(n/2) + n$
    b) $T(n) = 4\,T(n/2) + n^2$
    c) $T(n) = 9\,T(n/3) + n^3$
    d) $T(n) = 3\,T(n/4) + n \log n$
    e) $T(n) = 2\,T(n/4) + \sqrt{n}$
    f) $T(n) = 2\,T(2n/3) + 1$
    g) $T(n) = 4\,T(n/2) + n^2 / \log n$